

staggered adoption designs, as outside of this case it is more difficult to characterize those cells. Our characterization relies on the fact that for every $t \in \{2, \dots, T\}$, $\varepsilon_{fd,g,t} = D_{g,t} - D_{g,t-1} - (D_{.,t} - D_{.,t-1})$. $\varepsilon_{fd,g,t}$ is the difference between the change of the treatment in group g between $t-1$ and t , and the average change of the treatment across all groups.

Proposition 2 *Suppose that Assumptions 1-2 and 6 hold and for all g , $N_{g,t}$ does not depend on t . Then, for all (g, t) such that $D_{g,t} = 1$, $w_{fd,g,t} < 0$ if and only if $D_{g,t-1} = 1$ and $D_{.,t} - D_{.,t-1} > D_{.,t+1} - D_{.,t}$ (with the convention that $D_{.,T+1} = D_{.,T}$).*

Proposition 2 shows that for all $t \in \{2, \dots, T-1\}$ such that the increase in the proportion of treated units is larger from $t-1$ to t than from t to $t+1$, the period- t ATE of groups already treated in $t-1$ receives a negative weight. Moreover, if, at period T , at least one group becomes treated, the ATE of groups already treated in $T-1$ also receives a negative weight. Therefore, the treatment effect arising at the date when a group starts receiving the treatment does not receive a negative weight, only long-run treatment effects do. Then, negative weights are a concern when instantaneous and long-run treatment effects may differ. Proposition 2 also shows that the prevalence of negative weights depends on how the number of groups that start receiving the treatment at date t evolves with t . Assume for instance that this number decreases with t : many groups start receiving the treatment at date 1, a bit less start at date 2, etc., a case hereafter referred to as the “more early adopters” case. Then, if $N_{g,t}$ is constant across (g, t) , $D_{.,t} - D_{.,t-1}$ is decreasing in t , and all the long-run treatment effects receive negative weights, except maybe those of period T if $D_{.,T} = D_{.,T-1}$. Conversely, assume that the number of groups that start receiving the treatment at date t increases with t : few groups start receiving the treatment at date 1, a bit more start at date 2, etc., a case hereafter referred to as the “more late adopters” case. Then, if $N_{g,t}$ is constant across (g, t) , $D_{.,t} - D_{.,t-1}$ is increasing in t , and only the period- T long-run treatment effects receive negative weights. Overall, negative weights are much more prevalent in the “more early adopters” than in the “more late adopters” case.

We now come back to general sharp designs where the treatment may not follow a staggered adoption. Let $\tilde{\beta}_{fd} = E\left(\hat{\beta}_{fd} \middle| \mathbf{D}\right)$ denote the expectation of $\hat{\beta}_{fd}$ conditional on the vector of treatment assignments $\mathbf{D}$. Just as for $\tilde{\beta}_{fe}$, one can show that the minimal value of $\sigma(\tilde{\Delta})$ compatible with $\tilde{\beta}_{fd}$ and $\tilde{\Delta}^{TR} = 0$ is $\underline{\sigma}_{fd} = |\tilde{\beta}_{fd}|/\sigma(\mathbf{w}_{fd})$, where

$$\sigma(\mathbf{w}_{fd}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{fd,g,t} - 1)^2 \right)^{1/2}$$

is the standard deviation of the $\mathbf{w}_{fd}$ -weights. One can also show that $\underline{\sigma}_{fd}$, the minimal value of $\sigma(\tilde{\Delta})$ compatible with $\tilde{\beta}_{fd}$ and $\tilde{\Delta}_{g,t}$ of a different sign than $\tilde{\beta}_{fd}$ for all (g, t) , has the same expression as $\underline{\sigma}_{fe}$, except that one needs to replace the weights $w_{g,t}$ by the weights $w_{fd,g,t}$ in its

definition. Estimators of $\underline{\sigma}_{fe}$ and $\underline{\sigma}_{fd}$ (or $\underline{\sigma}_{fe}$ and $\underline{\sigma}_{fd}$) can then be used to determine which of $\widehat{\beta}_{fe}$ or $\widehat{\beta}_{fd}$ is more robust to heterogeneous treatment effects.

Finally, and similarly to the result shown in Corollary 2 for β_{fe} , β_{fd} is equal to δ^{TR} under common trends and the following assumption:

Assumption 8 (*w_{fd} uncorrelated with $\widetilde{\Delta}$*) $E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{fd,g,t} - 1) (\Delta_{g,t} - \Delta^{TR}) \right] = 0$.

Note that under the common trends assumption, one can jointly test Assumption 8 and Assumption 7, the assumption that the weights attached to β_{fe} are uncorrelated with the $\Delta_{g,t}$ s: if $\widehat{\beta}_{fe}$ and $\widehat{\beta}_{fd}$ are significantly different, at least one of these two assumptions must fail. In the second application we revisit in Section 6, $\widehat{\beta}_{fe}$ and $\widehat{\beta}_{fd}$ are significantly different.

4 An alternative estimator

In this section, we show that it is possible to estimate a well-defined causal effect even if treatment effects are heterogeneous across groups or over time. Let

$$\delta^S = E \left[\frac{1}{N_S} \sum_{(i,g,t): t \geq 2, D_{g,t} \neq D_{g,t-1}} [Y_{i,g,t}(1) - Y_{i,g,t}(0)] \right],$$

with $N_S = \sum_{(g,t): t \geq 2, D_{g,t} \neq D_{g,t-1}} N_{g,t}$. δ^S is the ATE of all switching cells. In staggered adoption designs, δ^S is the average of the treatment effect at the time when a group starts receiving the treatment, across all groups that become treated at some point.

We now show that δ^S can be unbiasedly estimated by a weighted average of DID estimators. This result holds under the following supplementary assumptions.

Assumption 9 (*Strong exogeneity for $Y(1)$*) For all $(g, t) \in \{1, \dots, G\} \times \{2, \dots, T\}$, $E(Y_{g,t}(1) - Y_{g,t-1}(1) | D_{g,1}, \dots, D_{g,T}) = E(Y_{g,t}(1) - Y_{g,t-1}(1))$.

Assumption 9 is the equivalent of Assumption 4, for the potential outcome with treatment. It requires that the shocks affecting a group's $Y_{g,t}(1)$ be mean independent of that group's treatment sequence.

Assumption 10 (*Common trends for $Y(1)$*) For $t \geq 2$, $E(Y_{g,t}(1) - Y_{g,t-1}(1))$ does not vary across g .

Again, Assumption 10 is the equivalent of Assumption 5, for the potential outcome with treatment. It requires that between each pair of consecutive periods, the expectation of the outcome with treatment follow the same evolution over time in every group. Assumptions 9 and 10 ensure that one can reconstruct the potential outcome that groups leaving the treatment between

$t - 1$ and t would have experienced if they had remained treated. In staggered adoption designs, Assumption 9 and 10 are not necessary for identification, because no group leaves the treatment. Together, Assumptions 5 and 10 imply that the ATE follows the same evolution over time in every group: $E(\Delta_{g,t}) = \eta_t + \theta_g$.¹³ This still allows for heterogeneous treatment effects across groups and over time.¹⁴

Assumption 11 (*Existence of “stable” groups*) For all $t \in \{2, \dots, T\}$:

1. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 0$ and $D_{g,t} = 1$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 0$.
2. If there is at least one $g \in \{1, \dots, G\}$ such that $D_{g,t-1} = 1, D_{g,t} = 0$, then there exists at least one $g' \neq g, g' \in \{1, \dots, G\}$ such that $D_{g',t-1} = D_{g',t} = 1$.

The first point of the stable groups assumption requires that between each pair of consecutive time periods, if there is a “joiner” (i.e., a group switching from being untreated to treated), then there should be another group that is untreated at both dates. The second point requires that between each pair of consecutive time periods, if there is a “leaver” (i.e., a group switching from being treated to untreated), then there should be another group that is treated at both dates.

Notice that under Assumption 11, groups’ treatments are not independent, so Assumption 3 cannot hold. Accordingly, we replace Assumption 3 by Assumption 12 below. Assumption 12 requires that conditional on its own treatments, a group’s outcomes be mean independent of the other groups’ treatments. It is weaker than Assumption 3. Assumption 11 is necessary to show that our estimator is unbiased, but it is not necessary to show that it is consistent. Accordingly, in Section ?? of the Web Appendix, we show that our estimator is consistent under Assumption 3. For every $g \in \{1, \dots, G\}$, let $\mathbf{D}_g = (D_{1,g}, \dots, D_{T,g})$.

Assumption 12 (*Mean independence between a group’s outcome and other groups treatments*) For all g and t , $E(Y_{g,t}(0)|\mathbf{D}) = E(Y_{g,t}(0)|\mathbf{D}_g)$ and $E(Y_{g,t}(1)|\mathbf{D}) = E(Y_{g,t}(1)|\mathbf{D}_g)$.

We can now define our estimator. For all $t \in \{2, \dots, T\}$ and for all $(d, d') \in \{0, 1\}^2$, let

$$N_{d,d',t} = \sum_{g: D_{g,t}=d, D_{g,t-1}=d'} N_{g,t} \quad (3)$$

¹³ It should be possible to weaken Assumptions 9-10, in particular to account for dynamic effects where $\Delta_{g,t}$ may depend on $(D_{g,1}, \dots, D_{g,t-1})$. This would however introduce complications that are beyond the scope of this paper.

¹⁴ Imposing Assumptions 9 and 10 does not change the decompositions obtained in Theorems 1 and 2. $Y_{g,t}(1)$ is observed for all the treated (g, t) cells entering these decompositions, so those assumptions do not bring identifying information for those cells.

denote the number of observations with treatment d' at period $t - 1$ and d at period t . Let

$$\begin{aligned} \text{DID}_{+,t} &= \sum_{g:D_{g,t}=1, D_{g,t-1}=0} \frac{N_{g,t}}{N_{1,0,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=D_{g,t-1}=0} \frac{N_{g,t}}{N_{0,0,t}} (Y_{g,t} - Y_{g,t-1}), \\ \text{DID}_{-,t} &= \sum_{g:D_{g,t}=D_{g,t-1}=1} \frac{N_{g,t}}{N_{1,1,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=0, D_{g,t-1}=1} \frac{N_{g,t}}{N_{0,1,t}} (Y_{g,t} - Y_{g,t-1}). \end{aligned}$$

Note that $\text{DID}_{+,t}$ is not defined when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 0$, or no group such that $D_{g,t} = 0, D_{g,t-1} = 0$. In such instances, we let $\text{DID}_{+,t} = 0$. Similarly, let $\text{DID}_{-,t} = 0$ when there is no group such that $D_{g,t} = 1, D_{g,t-1} = 1$ or no group such that $D_{g,t} = 0, D_{g,t-1} = 1$. Finally, let

$$\text{DID}_M = \sum_{t=2}^T \left(\frac{N_{1,0,t}}{N_S} \text{DID}_{+,t} + \frac{N_{0,1,t}}{N_S} \text{DID}_{-,t} \right).$$

Theorem 3 *If Assumptions 1, 2, 4, 5, and 9-12 hold, $E[\text{DID}_M] = \delta^S$.*

In Section ?? of the Web Appendix, we also show that when G goes to infinity, DID_M is a consistent and asymptotically normal estimator of δ^S . The DID_M estimator is computed by the `fuzzydid` and `did_multipligt` Stata packages.

Here is the intuition underlying Theorem 3. $\text{DID}_{+,t}$ compares the evolution of the mean outcome between $t - 1$ and t in two sets of groups: the joiners, and those remaining untreated. Under Assumptions 4 and 5, $\text{DID}_{+,t}$ estimates the joiners' treatment effect. Similarly, $\text{DID}_{-,t}$ compares the evolution of the outcome between $t - 1$ and t in two sets of groups: those remaining treated, and the leavers. Under Assumptions 9 and 10, it estimates the leavers' treatment effect. Finally, DID_M is a weighted average of those DIDs. Note that in staggered designs, there are no groups whose treatment decreases over time, so DID_M is only a weighted average of the $\text{DID}_{+,t}$ estimators. Note also that one can separately estimate the joiners' and the leavers' treatment effect, by computing separately weighted averages of the $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ estimators. The former estimator only relies on Assumptions 4 and 5, while the latter only relies on Assumptions 9 and 10.

DID_M is related to two other estimators. First, it is related to the Wald-TC estimator in point 2 of Theorem S1 in the Web Appendix of de Chaisemartin and D'Haultfœuille (2018), but the weighting of $\text{DID}_{+,t}$ and $\text{DID}_{-,t}$ therein differs. As a result, DID_M estimates Δ^S under weaker assumptions. DID_M is also related to the multi-period DID estimator in Imai and Kim (2018). However, the multi-period DID estimator is a weighted average of the $\text{DID}_{+,t}$, so it does not estimate the leavers' treatment effect, and applies to a smaller population. Besides, Imai and Kim (2018) do not establish the properties of their estimator. Finally, they do not generalize it to non-binary treatments, something we do in Section ?? of the Web Appendix.